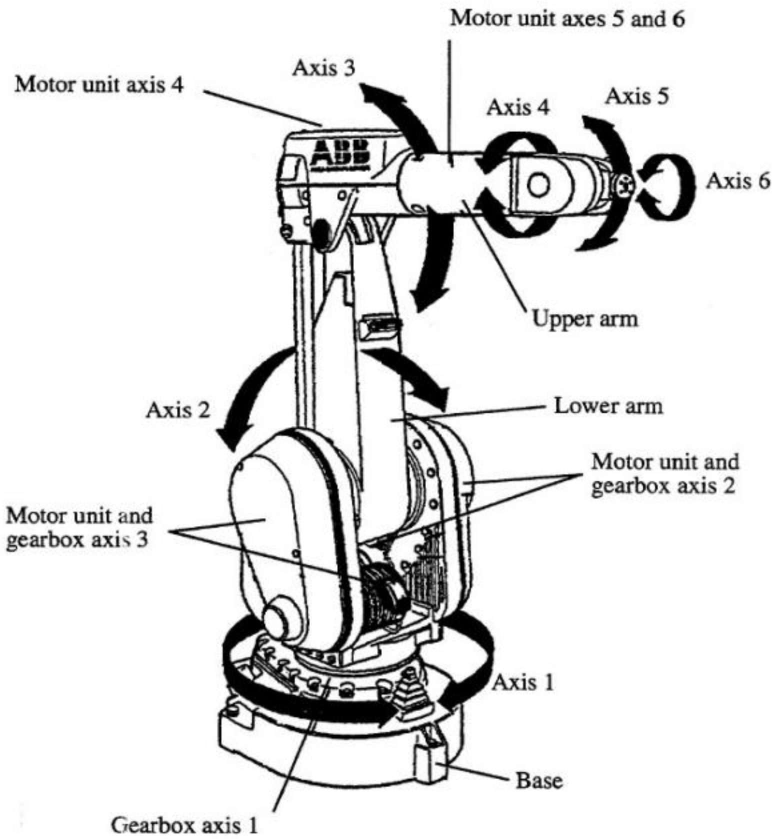

Figure 1: ABB IRB 1410 Industrial Robot.

Expt No:
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Date:
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SERIAL MANIPULATOR ROBOT (ABB IRB 1410)

1. AIM:

To understand the manual and programmed control of a 6-axis serial manipulator (ABB IRB 1410) for a pick-and-place operation using the ABB FlexPendant.

2. APPARATUS REQUIRED:

a) Robot: ABB IRB 1410 6-axis serial manipulator,
b) Controller: ABB IRC5 Controller,
c) Teach Pendant: ABB FlexPendant,
d) End Effector: Pneumatic Gripper,
e) Work Cell: Pegs and holder blocks, Safety interlocks.

3. THEORY / PRINCIPLE OF OPERATION:

The ABB IRB 1410 is a 6-axis serial manipulator. Its kinematic structure is an open chain of links connected by joints (like a human arm), where each joint adds a Degree of Freedom (DOF). It has 6 DOFs: the first three (J1-J3) generally position the wrist, and the final three (J4-J6) orient the end-effector.

a) Coordinate Systems:
   i) Joint Space: Manually controlling each joint (J1-J6) independently. This is non-intuitive for the tool's position but is the robot's native control space.
   ii) Cartesian/World Space: Commanding the Tool Center Point (TCP) to move in a straight line relative to the robot's fixed base (e.g., +X, +Y, +Z). The controller must solve the complex Inverse Kinematics (IK) equations to calculate the required angles for all 6 joints.
b) Programming:
   The method of "teaching" the robot by jogging it to a physical location and recording that point as a robtarget variable.
c) Motion Instructions (RAPID):
   i) MoveJ (Joint Move): The fastest move. All joints move from their start to end angles simultaneously. The TCP follows a non-linear, unpredictable arc. Used for large, non-critical "free space" movements.
   ii) MoveL (Linear Move): The TCP moves in a perfect straight line. The controller constantly re-calculates the IK solution. This is slower but essential for precise operations like approaching a part or welding.

4. PROCEDURE:

a) Jogging (Manual Control):
   i) Ensured the controller was in Manual Mode and gripped the FlexPendant, enabling the deadman switch.
   ii) From the ABB Main Menu, selected the Jogging window.
   iii) Selected "Axis 1-3" mode. Used the joystick to manually move Axis 1 (waist), Axis 2 (shoulder), and Axis 3 (elbow) individually.
   iv) Selected "Axis 4-6" mode. Used the joystick to manually move Axis 4 (wrist roll), Axis 5 (wrist pitch), and Axis 6 (flange roll).
   v) Switched the jogging coordinate system to "Linear" mode. Moved the joystick in the X, Y, and Z directions.
   vi) Switched the coordinate system to "Re-orient" mode and manipulated the joystick to change the gripper's orientation.
   vii) Using a combination of Linear and Re-orient modes, the Tool Center Point (TCP) was carefully jogged to the "pick" position above the peg.

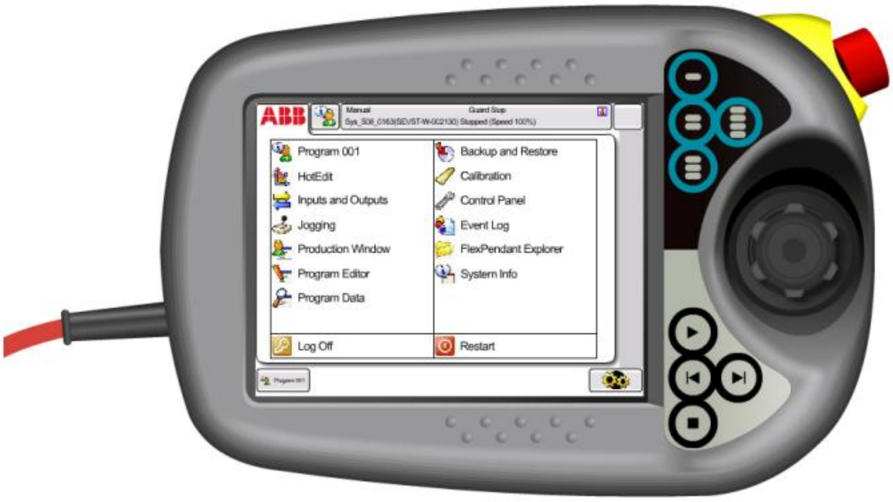

Figure 2: **ABB Flex Pendent.**

viii) From the Main Menu, selected Input and Output. Manually toggled the d01 and d03 digital outputs to close the gripper and secure the peg.
ix) Using "Axis" (joint) mode, the robot was jogged to a "safe" position high above the workspace.
x) Using "Linear" mode, the robot was jogged down to the "place" position.
xi) The Input and Output menu was used again to toggle the d01 and d03 bits to open the gripper, releasing the peg.

b) Programming (Automatic Control):
   i) From the Program Editor, a new MainModule was created with a PROC MAIN.
   ii) The Add Instruction command was used to select a MoveL (Linear) instruction.
   iii) The robot was jogged to a "Home" position, and the Modify Position button was pressed to "teach" this location, creating a robtarget variable p1.
   iv) Set instructions were added for the gripper I/O to ensure it was open (SET DO1 HIGH, SET D03 LOW).
   v) A MoveJ (Joint) instruction was added. The robot was jogged to an "approach" point above the peg, and the position was taught as p2.
   vi) A MoveL instruction was added. The robot was jogged slowly to the final "pick" position, and the position was taught as p3.
   vii) Set instructions were added to close the gripper (SET DO1 LOW, SET D03 HIGH).
   viii) Another MoveL was added to lift the peg straight up, taught as p4.
   ix) This process was repeated to teach an "approach_drop" point (p5), a "drop" point (p6), and the I/O commands to open the gripper.
   x) The program was concluded with a MoveJ back to the p1 "Home" position.
   xi) The Debug tool was used to check for syntax errors. The Program Pointer (PP) was set to Main (Check simultaneously in the Pendent).
   xii) In Manual Mode at 50% speed, the program was executed step-by-step and then run continuously.

5. **RESULT:**

The manual jogging and programmed pick-and-place operations were successfully studied. The experiment provided a clear, practical understanding of the difference between joint-space (MoveJ) and linear-space (MoveL) motion, the "teach" programming method, and the integration of I/O (gripper) commands within an automated sequence on a 6-axis serial manipulator.